

Fig. 2: EHT generator

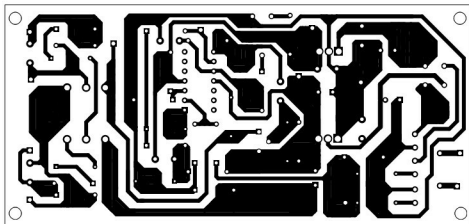


Fig. 3: An actual-size, single-side PCB for the pulse generator

transistor SL100 (T1) and a few discrete components.

Transformer X1 is a special type of step-down transformer that has two secondary windings, viz. 20V-0-20V, 500 mA and 18V-0-18V, 500 mA, respectively. The power supply for circuit operation is built using 20V-0-20V secondary winding of transformer X1, followed by rectifier BR1. Capacitors C7 and C8 are used to filter positive and negative ripples, respectively. IC2 is a +15V regulator, while IC3 is a -15V regulator. Capacitors C9, C10, C11 and C12 are decoupling capacitors used to decouple DC power supplies for high-frequency noise, if any, present on the DC power supply line.

Working of the pulse generator section is simple. When mains power is applied to the primary of X1, regulators IC2 and IC3 provide +15 V and -15

V, respectively, for circuit operation. The 18V-0-18V secondary of transformer X1 is rectified by diodes D1 and D2 to provide full rectified wave signal at test point TP3. This rectified wave after level-dropping by resistors R2 and R3 is fed to the inverting input of op-amp A1, where it is compared with a fixed voltage level, set by resistors R4 and R5, to produce rectangular pulses at the output (OUT1).

These pulses are converted into sawtooth wave by an arrangement made using R7, R8, C1 and ZD1. The sawtooth wave is further fed to the inverting terminal of op-amp A3 through resistor R9, where it is compared with a fixed DC voltage level, adjustable via VR1.

Depending upon the setting of VR1, the output at pin 8 of IC1 also varies. The trigger pulse seen at the collector of transistor T1 will also

move with the DC level set by VR1, since this is a result of sawtooth wave's comparison with the DC voltage.

Once you get sharp pulses at the collector of transistor T1, you will get two pulses at G1K1 and G2K2 generated by the pulse transformer (X3). These pulses have to be positive so that they can trigger the two SCRs connected back to back. These pulses are fairly sharp but still have sufficient time intervals (a few milliseconds), to make triggering possible.

The pulses at G1K1 and G2K2 (shown in Fig. 1) are used to trigger the SCRs as shown in Fig. 2. The turn ratio of EHT transformer X2 is $N2/N1=25$. At the secondary of X2, ten units of resistor and capacitor, as shown in the circuit, are connected in series for transient suppression. The R18-C13 and R20-C14 combinations are used as snubber circuits. Resistor R19 is used to eliminate phase lag.

To develop a high DC voltage, ten rectifier diodes (BY255 or equivalent), each shunted by a 1-mega-ohm resistor and 0.1µF capacitor, are used in each branch of the rectifier bridge connected across the high-voltage secondary. The same is indicated in Fig. 2 by a block for simplicity. The output is further passed through the standard bleeder and filter blocks to achieve the final EHT voltage output.

Note that for suppressing the